

HACKOUT'26 — PHASE 1 IDEATION + IMPLEMENTED MVP

PLUGIN

EV Charging Network Renewable-Optimization Platform

Charge at the right station, at the right time, with the cleanest practical energy.

The problem

EV charging demand can concentrate during expensive, carbon-intensive grid peaks even when renewable supply is expected later. Drivers need simple choices, station operators need capacity guidance, and grid operators need an aggregated view of risk.

Our solution

PLUGIN preserves the existing station discovery, booking, session, wallet, billing, and security flows, then adds an explainable renewable-intelligence layer. It forecasts regional energy conditions and EV demand, compares feasible charging windows, and communicates greener, cheaper, faster, and balanced choices.

Why it is credible for a hackathon

- Works offline with deterministic India-oriented DEMO/SIMULATED data, so judging is repeatable.
- Supports a normalized external provider through environment configuration, caching, validation, timeouts, and fallback.
- Labels estimates honestly; it does not present simulated values as utility-grade telemetry.
- Uses role-based access for customers, station operators, and administrators.

Existing EV platform = foundation | Renewable intelligence = optimization layer

How the app works

One end-to-end flow from regional forecast to an explainable charging choice

1 Forecast	2 Request	3 Evaluate	4 Recommend	5 Operate
Normalize solar, wind, demand, price, and carbon signals.	Driver enters energy, power, station capacity, and time window.	Score feasible 30-minute schedules within hard constraints.	Return four ranked strategies with cost and carbon impact.	Show station capacity, surplus, demand, and peak-risk windows.

Decision flow

1. The public energy screen requests the current regional outlook and forecast.
2. If an external provider is configured, the server requests its normalized payload with strict timeouts and numeric validation. Otherwise it uses the local deterministic model.
3. Provider results are cached to reduce calls and protect quotas. Any provider failure falls back safely to visibly labeled demo data.
4. An authenticated customer submits a flexible charging request. The backend rejects expired, infeasible, or longer-than-48-hour windows.
5. The optimizer limits charging power to both vehicle/charger power and available station capacity, then estimates delivered energy, renewable share, tariff, grid load, and carbon for every candidate.
6. Four alternatives are returned. The selected preference is shown first, but the driver can compare all trade-offs before using the existing booking flow.
7. Operators review a 24-hour demand and renewable outlook to identify surplus windows, capacity pressure, and possible peaks.

What the recommendation explains

Persona	What they do	Value delivered
Timing	Start and end of the suggested window	Makes the recommendation actionable
Energy	Required and deliverable kWh	Shows feasibility under capacity limits
Money	Estimated INR cost	Supports cheapest-versus-greenest comparison
Impact	Renewable share, carbon, carbon saved, green score	Makes sustainability understandable

Role journeys

The same forecast becomes a different decision for each stakeholder

Persona	What they do	Value delivered
Driver / customer	Select region; view green outlook; enter charging need and flexibility; compare GREENEST, CHEAPEST, FASTEST, and BALANCED options.	A clear choice with estimated price, carbon, renewable share, score, and explanation.
Station operator	Review current load versus capacity, predicted EV demand, surplus periods, and peak-risk hours.	Move flexible demand away from congestion and prepare station capacity.
Administrator / grid persona	Review aggregated regional renewable supply, demand, overload risk, and forecast horizon.	Demonstrate how EV flexibility can support regional grid planning.

Customer journey

Discover station → View energy outlook → Compare charging options → Continue existing booking/session/payment flow

The MVP deliberately recommends rather than silently changing a booking. This keeps the new layer explainable and prevents unexpected scheduling or billing behavior.

Operator journey

Open Grid Energy → Check capacity → Find renewable surplus → Identify peak risk → Guide flexible demand

Station operators retain their existing operational role. Administrators represent the grid-operator perspective in the hackathon build; a separate production grid role requires a full authorization migration.

Core functional requirements met

- Public, region-aware current and forecast energy views.
- Deterministic offline demonstration and explicit data-quality labels.
- Safe external-provider fallback and request caching.
- Capacity-aware optimization with bounded inputs and explainable outcomes.
- Protected operator and administrator dashboards.
- Responsive web flow and a mobile renewable-energy screen.

Implementation architecture

A small, testable layer added to the existing monorepo

1	2	3	4
Web / mobile	Secure API	Services	Data source
Energy screen and smart options form	Public energy + protected optimization/operator routes	Provider selection, cache, demand model, optimizer	External normalized adapter or DEMO fallback

API surface

Persona	What they do	Value delivered
Public	GET /api/energy/current; GET /api/energy/forecast	Current regional conditions and forecast
Customer	POST /api/optimization/charging-options	Four feasible charging strategies
Operator	GET /api/operator/energy/dashboard	Station capacity and 24-hour operational outlook
Administrator	GET /api/grid/dashboard	Regional grid-persona overview

Data and quota safety

- Provider credentials are read from environment variables only and are never sent to web or mobile clients.
- Forecast responses are cached, with a minimum cache interval, to avoid unnecessary API usage.
- Connect/read timeouts, numeric bounds, maximum 48-hour horizons, and deterministic fallback prevent fragile demos.
- The external adapter expects a separately licensed normalized source; no unsupported government or utility API is claimed.

Optimization method

The current model is deterministic and explainable—not machine learning. It generates 30-minute candidates, enforces energy-delivery and power constraints, normalizes renewable, price, peak-load, carbon, and wait-time signals, and ranks them according to the requested preference. Equal inputs within the same forecast window produce repeatable results.

Validation and judge demo

What is proven today and what remains production work

Verified evidence

Persona	What they do	Value delivered
Backend	mvn clean verify	137 tests passed; Spring Boot production JAR built
Web	Security checks + production Vite build	Passed
Mobile	Security checks + Expo compatibility + Android export	Passed
Documents	DOCX/PDF render and page inspection	Submission artifacts verified

Five-minute demonstration

1. Open Green Energy and point out the DEMO/SIMULATED badge, regional selector, renewable mix, price, carbon, and next-hours chart.
2. Sign in as a customer, enter a flexible window, and compare the four strategies. Explain why GREENEST and CHEAPEST can choose different times.
3. Open the station-operator dashboard and show capacity, renewable surplus, demand forecast, and peak-risk timing.
4. Open the administrator grid view and explain how aggregated EV flexibility can reduce peak stress.
5. Disconnect external data or leave it unconfigured; demonstrate that the same labeled offline scenario remains available.

Honest MVP boundaries

- No physical grid dispatch, utility settlement, OCPP charger control, or certified carbon accounting.
- Recommendations do not yet automatically reschedule bookings or alter authoritative bills.
- Production rollout needs a licensed/approved live data source, regional calibration, monitoring, and operator agreements.
- Mobile dependency audit findings are primarily in the Expo/Metro toolchain; forcing the suggested major upgrade would be a separate compatibility project.

Outcome: a demonstrable, secure, explainable renewable-aware extension—not a replacement for the proven EV platform.